# 1. Business case : Predict the person will donate blood or not.

This data set is about blood donation prediction.This dataset contains blood donation details of regular university visits. The goal is to predict whether the person has donated blood in March 2007. The target is to predict whether a donor will give blood the next time in the blood donation campaign held at the university campus.

## Problem Statement:

- Create a predictive model which will help to repeat blood donations amongst donors based on a limited number of attributes.
- Prepare a complete data analysis report on the given data.

## Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as stats
import warnings
warnings.filterwarnings('ignore')

#loading datasets
data=pd.read_csv('Warm_Up_Predict_Blood_Donations_-_Traning_Data.csv')
data
```

```
     Unnamed: 0  Months since Last Donation  Number of Donations  \
0           619                           2                   50
1           664                           0                   13
2           441                           1                   16
3           160                           2                   20
4           358                           1                   24
..          ...                         ...                  ...
571         698                          23                    1
572         433                          16                    3
573         360                          21                    2
574         541                          39                    1
575          74                          72                    1

     Total Volume Donated (c.c.)  Months since First Donation  \
0                          12500                            98
1                           3250                            28
2                           4000                            35
3                           5000                            45
4                           6000                            77
..                           ...                           ...
571                          250                            23
572                          750                            86
```

```
573                          500                                52
574                          250                                39
575                          250                                72

     Made Donation in March 2007
0                              1
1                              1
2                              1
3                              1
4                              0
..                           ...
571                            0
572                            0
573                            0
574                            0
575                            0

[576 rows x 6 columns]
```

## DOMAIN ANALYSIS :

- 1.) Months since Last Donation: this is the number of months since this donor's most recent donation.
- 2.) Number of Donations : this is the total number of donations that the donor has made.
- 3.) Total Volume Donated : this is the total amound of blood that the donor has donated in cubuc centimeters.
- 4.) Months since First Donation : this is the number of months since the donor's first donation
- 5.) Made Donation in March 2007 : a binary variable representing whether he/she donated blood in March 2007 (1 stand for
donating blood; 0 stands for not donating blood).

### Basic Checks

```
data.head()

   Unnamed: 0  Months since Last Donation  Number of Donations  \
0         619                           2                   50
1         664                           0                   13
2         441                           1                   16
3         160                           2                   20
4         358                           1                   24

   Total Volume Donated (c.c.)  Months since First Donation  \
0                        12500                           98
1                         3250                           28
2                         4000                           35
3                         5000                           45
4                         6000                           77
```

```
   Made Donation in March 2007
0                            1
1                            1
2                            1
3                            1
4                            0
```

```
data.tail()
```

```
     Unnamed: 0  Months since Last Donation  Number of Donations  \
571         698                          23                    1
572         433                          16                    3
573         360                          21                    2
574         541                          39                    1
575          74                          72                    1

     Total Volume Donated (c.c.)  Months since First Donation  \
571                          250                            23
572                          750                            86
573                          500                            52
574                          250                            39
575                          250                            72

     Made Donation in March 2007
571                            0
572                            0
573                            0
574                            0
575                            0
```

```
data.shape
```

```
(576, 6)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 576 entries, 0 to 575
Data columns (total 6 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Unnamed: 0                   576 non-null    int64
 1   Months since Last Donation   576 non-null    int64
 2   Number of Donations          576 non-null    int64
 3   Total Volume Donated (c.c.)  576 non-null    int64
 4   Months since First Donation  576 non-null    int64
 5   Made Donation in March 2007  576 non-null    int64
dtypes: int64(6)
memory usage: 27.1 KB
```

- There's no null value.

```
data.describe()
```

```
       Unnamed: 0  Months since Last Donation  Number of Donations  \
count  576.000000                  576.000000           576.000000
mean   374.034722                    9.439236             5.427083
std    216.947773                    8.175454             5.740010
min      0.000000                    0.000000             1.000000
25%    183.750000                    2.000000             2.000000
50%    375.500000                    7.000000             4.000000
75%    562.500000                   14.000000             7.000000
max    747.000000                   74.000000            50.000000

       Total Volume Donated (c.c.)  Months since First Donation  \
count                   576.000000                   576.000000
mean                   1356.770833                    34.050347
std                    1435.002556                    24.227672
min                     250.000000                     2.000000
25%                     500.000000                    16.000000
50%                    1000.000000                    28.000000
75%                    1750.000000                    49.250000
max                   12500.000000                    98.000000

       Made Donation in March 2007
count                   576.000000
mean                      0.239583
std                       0.427200
min                       0.000000
25%                       0.000000
50%                       0.000000
75%                       0.000000
max                       1.000000
```

```
data.duplicated().sum()
```

```
0
```

# Vizualization

1.) Univariate Analysis :-

```
import sweetviz as sv
my_repo=sv.analyze(data)
my_repo.show_html('myrepo.html')
```

{"model_id":"3c0f87779b3046888807fa979f3a2e2a","version_major":2,"version_minor":0}

```
Report myrepo.html was generated! NOTEBOOK/COLAB USERS: the web
browser MAY not pop up, regardless, the report IS saved in your
notebook/colab files.
```

Insights :-

- Months Last Donation ranges from 0 to 74.
- Two values more than 70
- Total donation ranges from 1 to 50.
- There are 4 values more than 40 in total donation.
- There are 80% donors who have total volume donation count is between 250 c.c. to 2000 c.c.
- There are 12 donors who have more than 90 months since first blood donation.
- There are 438 obsevations are lie in negative class.

## 2.) Bivariate Analysis

```
data.columns
```

```
Index(['Unnamed: 0', 'Months since Last Donation', 'Number of
Donations',
       'Total Volume Donated (c.c.)', 'Months since First Donation',
       'Made Donation in March 2007'],
      dtype='object')
```

```
data.rename({'Made Donation in March 2007' :
'Made_Donation_in_March_2007'},axis=1,inplace=True)
```

```
data
```

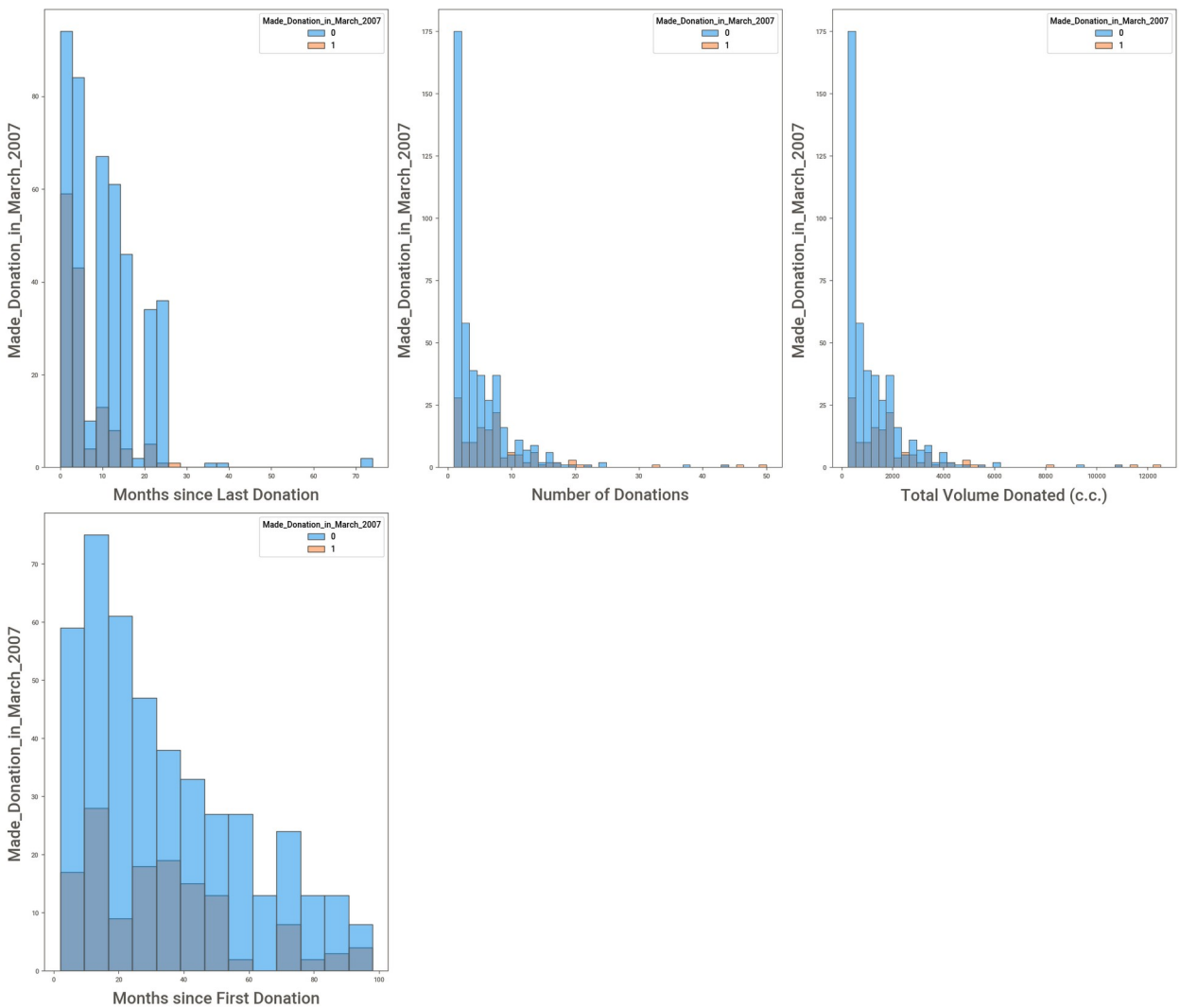
```
     Unnamed: 0  Months since Last Donation  Number of Donations  \
0           619                           2                   50
1           664                           0                   13
2           441                           1                   16
3           160                           2                   20
4           358                           1                   24
..          ...                         ...                  ...
571         698                          23                    1
572         433                          16                    3
573         360                          21                    2
574         541                          39                    1
575          74                          72                    1

     Total Volume Donated (c.c.)  Months since First Donation  \
0                          12500                           98
1                           3250                           28
2                           4000                           35
3                           5000                           45
4                           6000                           77
..                           ...                          ...
571                          250                           23
572                          750                           86
573                          500                           52
574                          250                           39
```

```
575                           250                            72

     Made_Donation_in_March_2007
0                              1
1                              1
2                              1
3                              1
4                              0
..                           ...
571                            0
572                            0
573                            0
574                            0
575                            0

[576 rows x 6 columns]
```

```
data1 = data[['Months since Last Donation', 'Number of Donations',
       'Total Volume Donated (c.c.)', 'Months since First Donation']]
data1
```

```
     Months since Last Donation  Number of Donations  \
0                             2                   50
1                             0                   13
2                             1                   16
3                             2                   20
4                             1                   24
..                          ...                  ...
571                          23                    1
572                          16                    3
573                          21                    2
574                          39                    1
575                          72                    1

     Total Volume Donated (c.c.)  Months since First Donation
0                          12500                           98
1                           3250                           28
2                           4000                           35
3                           5000                           45
4                           6000                           77
..                           ...                          ...
571                          250                           23
572                          750                           86
573                          500                           52
574                          250                           39
575                          250                           72

[576 rows x 4 columns]
```

```
plt.figure(figsize=(20,25))
plotnumber = 1

for column in data1:
    if plotnumber<=9 :
        ax = plt.subplot(3,3,plotnumber)

sns.histplot(x=data1[column],hue=data.Made_Donation_in_March_2007)
        plt.xlabel(column,fontsize=20,labelpad=10)

plt.ylabel('Made_Donation_in_March_2007',fontsize=20,labelpad=10)
    plotnumber+=1
plt.tight_layout()
```




Insights :-

- The number of donors who could not donate blood is more than those who donated blood 10 months ago and still can donate blood in March 2007.

- There is not a single donor whose last blood donation is between 40 to 70 months.
- Who has done blood donation 2 times and in March 2007 also there are around 30 donors who can donate blood and around 175 who can not donate blood.
- Donors whose number of donation is more than 45 times can donate blood in March-2007.
- Donors whose blood donut count is more than 15000 cc can donate blood in March-2007.
- The number of donors whose first donation is between 25 to 50 months is high and they can do it even in March-2007.

## Data Preprocessing

```
import pandas as pd
import numpy as np
from scipy import stats
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

df=pd.read_csv('Warm_Up_Predict_Blood_Donations_-_Traning_Data.csv')

df
```

```
     Unnamed: 0  Months since Last Donation  Number of Donations  \
0           619                           2                   50
1           664                           0                   13
2           441                           1                   16
3           160                           2                   20
4           358                           1                   24
..          ...                         ...                  ...
571         698                          23                    1
572         433                          16                    3
573         360                          21                    2
574         541                          39                    1
575          74                          72                    1

     Total Volume Donated (c.c.)  Months since First Donation  \
0                          12500                           98
1                           3250                           28
2                           4000                           35
3                           5000                           45
4                           6000                           77
..                           ...                          ...
571                          250                           23
572                          750                           86
573                          500                           52
574                          250                           39
575                          250                           72
```

```
     Made Donation in March 2007
0                              1
1                              1
2                              1
3                              1
4                              0
..                           ...
571                            0
572                            0
573                            0
574                            0
575                            0

[576 rows x 6 columns]
```

```
df.info()       ## Checking for null values
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 576 entries, 0 to 575
Data columns (total 6 columns):
 #   Column                       Non-Null Count  Dtype
---  ------                       --------------  -----
 0   Unnamed: 0                   576 non-null    int64
 1   Months since Last Donation   576 non-null    int64
 2   Number of Donations          576 non-null    int64
 3   Total Volume Donated (c.c.)  576 non-null    int64
 4   Months since First Donation  576 non-null    int64
 5   Made Donation in March 2007  576 non-null    int64
dtypes: int64(6)
memory usage: 27.1 KB
```

```
df.describe()
```

```
       Unnamed: 0  Months since Last Donation  Number of Donations  \
count  576.000000                  576.000000           576.000000
mean   374.034722                    9.439236             5.427083
std    216.947773                    8.175454             5.740010
min      0.000000                    0.000000             1.000000
25%    183.750000                    2.000000             2.000000
50%    375.500000                    7.000000             4.000000
75%    562.500000                   14.000000             7.000000
max    747.000000                   74.000000            50.000000

       Total Volume Donated (c.c.)  Months since First Donation  \
count                   576.000000                   576.000000
mean                   1356.770833                    34.050347
std                    1435.002556                    24.227672
min                     250.000000                     2.000000
25%                     500.000000                    16.000000
50%                    1000.000000                    28.000000
```

```
75%                     1750.000000                     49.250000
max                    12500.000000                     98.000000

       Made Donation in March 2007
count                   576.000000
mean                      0.239583
std                       0.427200
min                       0.000000
25%                       0.000000
50%                       0.000000
75%                       0.000000
max                       1.000000
```

```
df.duplicated().sum()
```

```
0
```

```
df.drop('Unnamed: 0',axis=1,inplace=True)    ### Dropping Unnamed: 0
```

```
df
```

```
     Months since Last Donation  Number of Donations  \
0                             2                   50
1                             0                   13
2                             1                   16
3                             2                   20
4                             1                   24
..                          ...                  ...
571                          23                    1
572                          16                    3
573                          21                    2
574                          39                    1
575                          72                    1

     Total Volume Donated (c.c.)  Months since First Donation  \
0                          12500                           98
1                           3250                           28
2                           4000                           35
3                           5000                           45
4                           6000                           77
..                           ...                          ...
571                          250                           23
572                          750                           86
573                          500                           52
574                          250                           39
575                          250                           72

     Made Donation in March 2007
0                              1
1                              1
2                              1
```

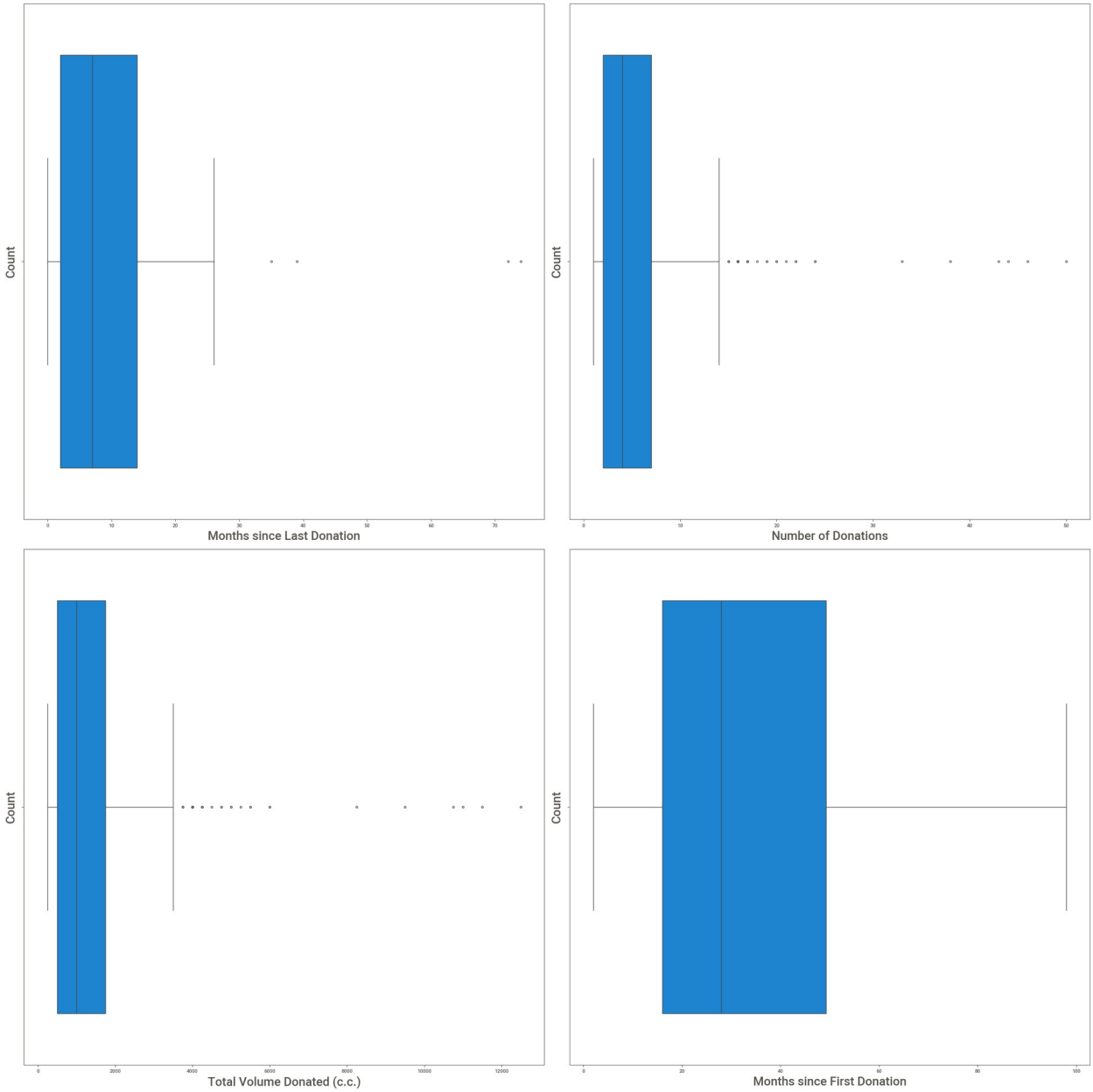
```
3                                 1
4                                 0
..                              ...
571                               0
572                               0
573                               0
574                               0
575                               0

[576 rows x 5 columns]
```

## Handling Outliers

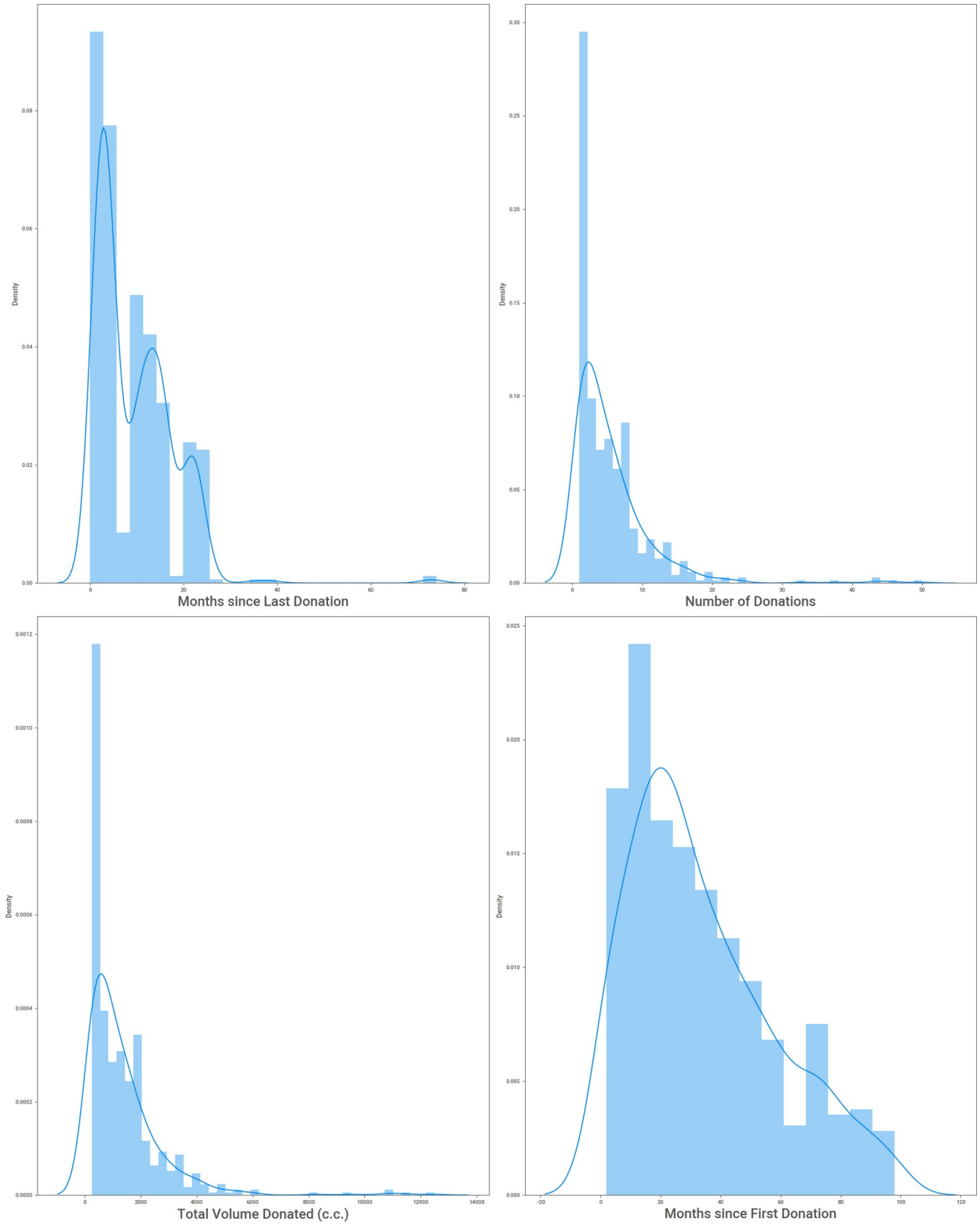
```
df.columns
```

```
Index(['Months since Last Donation', 'Number of Donations',
       'Total Volume Donated (c.c.)', 'Months since First Donation',
       'Made Donation in March 2007'],
      dtype='object')
```

```
df2=df[['Months since Last Donation', 'Number of Donations',
       'Total Volume Donated (c.c.)', 'Months since First Donation']]

# Checking for Outliers
plt.figure(figsize=(25,25),facecolor='white')
pltno=1
for column in df2:
    if pltno<=5:
        ax=plt.subplot(2,2,pltno)
        sns.boxplot(x=df2[column])
        plt.xlabel(column,fontsize=20)
        plt.ylabel('Count',fontsize=20)
    pltno+=1
plt.tight_layout()       ### Checking for outliers using Boxplot
```




```
plt.figure(figsize=(20,25), facecolor='white')
pltno=1
for column in df2:
    if pltno<=5:
        ax=plt.subplot(2,2,pltno)
        sns.distplot(x=df2[column])
        plt.xlabel(column,fontsize=20)
    pltno+=1
plt.tight_layout()        #Checking the Skewness of different columns
```




```
df=df.rename(columns={'Months since Last Donation':'ld', 'Number of
Donations':'Number',
```

```
        'Total Volume Donated (c.c.)':'Volume', 'Months since First
Donation':'fd'})    ### renaming columns

df
```

```
     ld  Number  Volume  fd  Made Donation in March 2007
0     2      50   12500  98                            1
1     0      13    3250  28                            1
2     1      16    4000  35                            1
3     2      20    5000  45                            1
4     1      24    6000  77                            0
..   ..     ...     ...  ..                          ...
571  23       1     250  23                            0
572  16       3     750  86                            0
573  21       2     500  52                            0
574  39       1     250  39                            0
575  72       1     250  72                            0

[576 rows x 5 columns]
```

Last donation

```
iqr=stats.iqr(df.ld,interpolation='midpoint')
q1=df.ld.quantile(0.25)
q3=df.ld.quantile(0.75)
min_limit=q1-1.5*iqr
max_limit=q3+1.5*iqr
print('IQR =',iqr)
print('min_limit',min_limit)
print('max_limit',max_limit)
```

```
IQR = 12.0
min_limit -16.0
max_limit 32.0
```

```
df.loc[df['ld']<min_limit]      # Checking values less than min limit
```

```
Empty DataFrame
Columns: [ld, Number, Volume, fd, Made Donation in March 2007]
Index: []
```

```
df.loc[df['ld']>max_limit,'ld']=np.median(df.ld)
```

```
df.loc[df['ld']>max_limit]
```

```
Empty DataFrame
Columns: [ld, Number, Volume, fd, Made Donation in March 2007]
Index: []
```

Number

```
iqr=stats.iqr(df.Number,interpolation='midpoint')
q1=df.Number.quantile(0.25)
q3=df.Number.quantile(0.75)
min_limit=q1-1.5*iqr
max_limit=q3+1.5*iqr
print('IQR =',iqr)
print('min_limit',min_limit)
print('max_limit',max_limit)
```

```
IQR = 5.0
min_limit -5.5
max_limit 14.5
```

```
df.loc[df['Number']>max_limit]
```

| | ld | Number | Volume | fd | Made Donation in March 2007 |
|---|---|---|---|---|---|
| 0 | 2 | 50 | 12500 | 98 | 1 |
| 2 | 1 | 16 | 4000 | 35 | 1 |
| 3 | 2 | 20 | 5000 | 45 | 1 |
| 4 | 1 | 24 | 6000 | 77 | 0 |
| 8 | 5 | 46 | 11500 | 98 | 1 |
| 15 | 2 | 15 | 3750 | 49 | 1 |
| 28 | 2 | 16 | 4000 | 64 | 0 |
| 37 | 4 | 20 | 5000 | 69 | 1 |
| 47 | 4 | 19 | 4750 | 69 | 1 |
| 53 | 3 | 16 | 4000 | 74 | 0 |
| 60 | 4 | 17 | 4250 | 71 | 1 |
| 79 | 3 | 17 | 4250 | 86 | 0 |
| 84 | 6 | 17 | 4250 | 70 | 0 |
| 92 | 11 | 24 | 6000 | 64 | 0 |
| 188 | 11 | 22 | 5500 | 98 | 0 |
| 190 | 11 | 17 | 4250 | 79 | 1 |
| 220 | 14 | 18 | 4500 | 78 | 0 |
| 222 | 14 | 16 | 4000 | 70 | 0 |
| 254 | 14 | 16 | 4000 | 98 | 0 |
| 263 | 23 | 38 | 9500 | 98 | 0 |
| 281 | 23 | 15 | 3750 | 57 | 0 |
| 386 | 2 | 43 | 10750 | 86 | 1 |
| 387 | 6 | 22 | 5500 | 28 | 1 |
| 388 | 2 | 44 | 11000 | 98 | 0 |
| 390 | 2 | 21 | 5250 | 52 | 1 |
| 397 | 4 | 33 | 8250 | 98 | 1 |
| 432 | 4 | 16 | 4000 | 98 | 1 |
| 484 | 11 | 16 | 4000 | 89 | 0 |
| 503 | 16 | 16 | 4000 | 77 | 0 |
| 518 | 16 | 15 | 3750 | 87 | 0 |
| 523 | 23 | 19 | 4750 | 62 | 0 |

```
df.loc[df['Number']>max_limit,'Number']=np.median(df.Number)
df.loc[df['Number']>max_limit]
```

```
Empty DataFrame
Columns: [ld, Number, Volume, fd, Made Donation in March 2007]
Index: []
```

## FD

```
iqr=stats.iqr(df.fd,interpolation='midpoint')
q1=df.fd.quantile(0.25)
q3=df.fd.quantile(0.75)
min_limit=q1-1.5*iqr
max_limit=q3+1.5*iqr
print('IQR =',iqr)
print('min_limit',min_limit)
print('max_limit',max_limit)
```

```
IQR = 33.5
min_limit -34.25
max_limit 99.5
```

```
df.loc[df['fd']<min_limit]
```

```
Empty DataFrame
Columns: [ld, Number, Volume, fd, Made Donation in March 2007]
Index: []
```

```
df.loc[df['fd']>max_limit]
```

```
Empty DataFrame
Columns: [ld, Number, Volume, fd, Made Donation in March 2007]
Index: []
```

```
df.loc[df['fd']>max_limit,'fd']=np.median(df.fd)
df.loc[df['fd']>max_limit]
```

```
Empty DataFrame
Columns: [ld, Number, Volume, fd, Made Donation in March 2007]
Index: []
```

## Feature Engg

```
df.head()
```

```
   ld  Number  Volume  fd  Made Donation in March 2007
0   2       4   12500  98                            1
1   0      13    3250  28                            1
2   1       4    4000  35                            1
```

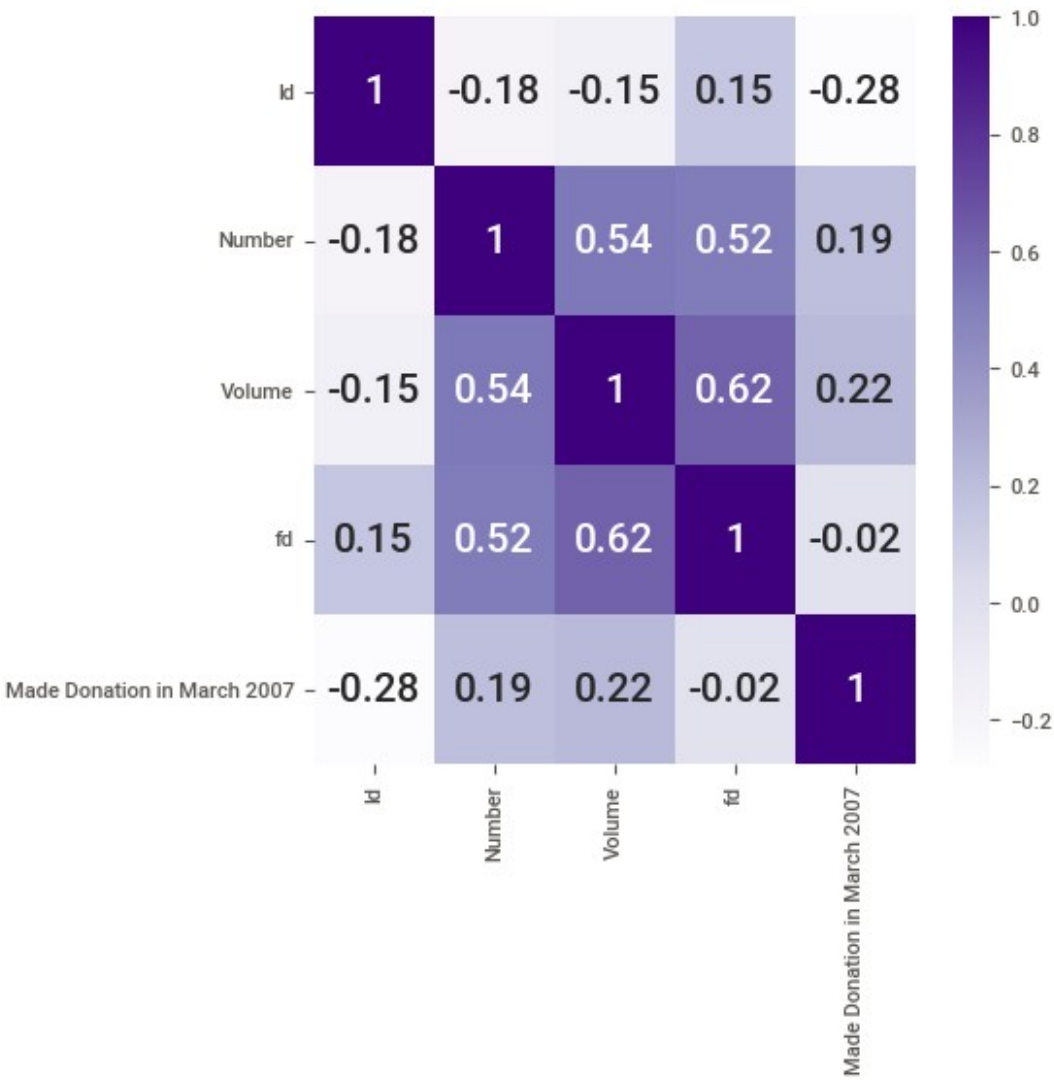
```
3   2       4    5000  45                              1
4   1       4    6000  77                              0
```

```
df.corr()
```

```
                                    ld    Number    Volume        fd  \
ld                           1.000000 -0.184635 -0.150572  0.153669
Number                      -0.184635  1.000000  0.536599  0.518697
Volume                      -0.150572  0.536599  1.000000  0.622116
fd                           0.153669  0.518697  0.622116  1.000000
Made Donation in March 2007 -0.275395  0.191408  0.220615 -0.019819
```

```
                              Made Donation in March 2007
ld                                              -0.275395
Number                                           0.191408
Volume                                           0.220615
fd                                              -0.019819
Made Donation in March 2007                      1.000000
```

```
plt.figure(figsize=(5,5))
sns.heatmap(df.corr(),annot=True,annot_kws={"size":15},cmap="Purples")
plt.show()
```




we will drop number as it is showing high corr.

```
df.drop(['Number'],axis=1,inplace=True)

df
```

| | ld | Volume | fd | Made Donation in March 2007 |
|---|---|---|---|---|
| 0 | 2 | 12500 | 98 | 1 |
| 1 | 0 | 3250 | 28 | 1 |
| 2 | 1 | 4000 | 35 | 1 |
| 3 | 2 | 5000 | 45 | 1 |
| 4 | 1 | 6000 | 77 | 0 |
| .. | .. | ... | .. | ... |
| 571 | 23 | 250 | 23 | 0 |
| 572 | 16 | 750 | 86 | 0 |
| 573 | 21 | 500 | 52 | 0 |
| 574 | 7 | 250 | 39 | 0 |

```
575   7     250  72                              0

[576 rows x 4 columns]
```

```
df=df.rename(columns={'Made Donation in March
2007':'Made_Donation_in_March_2007'})
```

# Model Creation

```
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.metrics import
accuracy_score,f1_score,confusion_matrix,classification_report
from sklearn.model_selection import RandomizedSearchCV
```

## Scaling

Now, we will normalize each numerical feature by using Standardscaler()

```
std=StandardScaler()

df[['ld', 'Volume','fd']] = std.fit_transform(df[['ld',
'Volume','fd']])
df.head()
```

```
         ld    Volume        fd  Made_Donation_in_March_2007
0 -1.005231  7.772052  2.641824                            1
1 -1.288160  1.320468 -0.249946                            1
2 -1.146695  1.843569  0.039231                            1
3 -1.005231  2.541038  0.452341                            1
4 -1.146695  3.238506  1.774293                            0
```

# Define X and y

```
x = df.drop('Made_Donation_in_March_2007',axis = 1)
y = df.Made_Donation_in_March_2007
x_train, x_test, y_train, y_test = train_test_split(x,y, test_size =
0.2, random_state = 42)
```

# Checking target data to see if it is balanced or not

```
y_train.value_counts()

Made_Donation_in_March_2007
0    351
```

```
1    109
Name: count, dtype: int64
```

- Data is not Balance

## Balancing the data using SMOTE

```
from imblearn.over_sampling import SMOTE

sm=SMOTE()
x_sm,y_sm=sm.fit_resample(x_train,y_train)

y_sm.value_counts()
```

```
Made_Donation_in_March_2007
0    351
1    351
Name: count, dtype: int64
```

# 1. Random Forest

```
from sklearn.ensemble import RandomForestClassifier
rf = RandomForestClassifier()
rf.fit(x_train,y_train)
```

```
RandomForestClassifier()
```

## Training

```
y_pred_train_rf = rf.predict(x_train)
a1=accuracy_score(y_train,y_pred_train_rf)
a1
```

```
0.9434782608695652
```

```
print(classification_report(y_train,y_pred_train_rf))
```

```
              precision    recall  f1-score   support

           0       0.95      0.98      0.96       351
           1       0.93      0.83      0.87       109

    accuracy                           0.94       460
   macro avg       0.94      0.90      0.92       460
weighted avg       0.94      0.94      0.94       460
```

## Testing

```
y_pred_test_rf = rf.predict(x_test)
b1=accuracy_score(y_test,y_pred_test_rf)
b1
```

```
0.7413793103448276
```

```
print(classification_report(y_test,y_pred_test_rf))
```

```
              precision    recall  f1-score   support

           0       0.81      0.85      0.83        87
           1       0.48      0.41      0.44        29

    accuracy                           0.74       116
   macro avg       0.65      0.63      0.64       116
weighted avg       0.73      0.74      0.73       116
```

```
rf1=RandomForestClassifier()
rf1.fit(x_sm,y_sm)
```

```
RandomForestClassifier()
```

```
y_pred=rf1.predict(x_test)

c1=accuracy_score(y_test,y_pred)
c1
```

```
0.7241379310344828
```

# Hyperparameter Tunning

```
rf2 = RandomForestClassifier( n_estimators=150,
    criterion='entropy',
    max_depth=20,
    min_samples_split=2,
    min_samples_leaf=2,
    min_weight_fraction_leaf=0.0,
    max_features='sqrt',
    max_leaf_nodes=None,
    min_impurity_decrease=0.0,
    bootstrap=True,
    oob_score=False,
    n_jobs=None,
    random_state=None,
    verbose=0,
    warm_start=False,
    class_weight=None,
    ccp_alpha=0.0,
    max_samples=None,)
```

```
rf2.fit(x_train,y_train)
```

```
RandomForestClassifier(criterion='entropy', max_depth=20,
min_samples_leaf=2,
                       n_estimators=150)
```

```
y_pred_rf2 = rf2.predict(x_test)
```

```
accuracy_score(y_test,y_pred_rf2)
```

```
0.7672413793103449
```

```
from sklearn.model_selection import RandomizedSearchCV

n_estimators = [int(x) for x in np.linspace(start=200, stop=2000,
num=10)]#List Comprehension-using for loop in list
max_features = ['auto', 'sqrt']#maximum number of features allowed to
try in individual tree
max_depth = [int(x) for x in np.linspace(10, 110, num=11)]#List
Comprehension-using for loop in list
max_depth.append(None)
min_samples_split = [2, 5, 10]#minimum number of samples required to
split an internal node
min_samples_leaf = [1, 2, 4]#minimum number of samples required to be
at a leaf node.
bootstrap = [True, False]#sampling

#dictionary for hyperparameters
random_grid = {'n_estimators': n_estimators, 'max_features':
max_features,
               'max_depth': max_depth, 'min_samples_split':
min_samples_split,
               'min_samples_leaf': min_samples_leaf, 'bootstrap':
bootstrap}

rf_clf1 = RandomForestClassifier(random_state=42)#model

rf_cv = RandomizedSearchCV(estimator=rf_clf1,
scoring='f1',param_distributions=random_grid, n_iter=100, cv=3,
                               verbose=2, random_state=42, n_jobs=-1)
#estimator--number of decision tree
#scoring--->performance matrix to check performance
#param_distribution-->hyperparametes(dictionary we created)
#n_iter--->Number of parameter settings that are sampled. n_iter
trades off runtime vs quality of the solution.default=10
##cv------> number of flods
#verbose=Controls the verbosity: the higher, the more messages.
#n_jobs---->Number of jobs to run in parallel,-1 means using all
processors.

rf_cv.fit(x_train, y_train)##training data on randomsearch cv
```

```
rf_best_params = rf_cv.best_params_##it will give you best parameters
print(f"Best paramters: {rf_best_params})")##printing  best parameters
```

```
Fitting 3 folds for each of 100 candidates, totalling 300 fits
Best paramters: {'n_estimators': 1000, 'min_samples_split': 10,
'min_samples_leaf': 1, 'max_features': 'sqrt', 'max_depth': 80,
'bootstrap': False})
```

```
d1=rf_best_params
d1
```

```
{'n_estimators': 1000,
 'min_samples_split': 10,
 'min_samples_leaf': 1,
 'max_features': 'sqrt',
 'max_depth': 80,
 'bootstrap': False}
```

```
rf_clf2 = RandomForestClassifier(n_estimators= 400, min_samples_split=
5, min_samples_leaf= 1, max_features= 'sqrt', max_depth= 30,
bootstrap= True)#passing best parameter to randomforest
rf_clf2.fit(x_train, y_train)#training
y_predict=rf_clf2.predict(x_test)#testing
```

```
e1=accuracy_score(y_test,y_predict)
e1
```

```
0.75
```

# SVM

```
from sklearn.svm import SVC
svc=SVC(
    C=1.0,
    kernel='rbf',
    degree=3,
    gamma='scale',
    coef0=0.0,
    shrinking=True,
    probability=False,
    tol=0.001,
    cache_size=200,
    class_weight=None,
    verbose=False,
    max_iter=-1,
    decision_function_shape='ovr',
    break_ties=False,
    random_state=None)
```

```
svc.fit(x_train,y_train)
```

```
SVC()
```

```
y_pred_train_svc=svc.predict(x_train)
```

```
a2=accuracy_score(y_train,y_pred_train_svc)
a2
```

```
0.7804347826086957
```

```
print(classification_report(y_train , y_pred_train_svc))
```

```
              precision    recall  f1-score   support

           0       0.78      0.99      0.87       351
           1       0.79      0.10      0.18       109

    accuracy                           0.78       460
   macro avg       0.78      0.55      0.53       460
weighted avg       0.78      0.78      0.71       460
```

```
print(confusion_matrix(y_train,y_pred_train_svc))
```

```
[[348   3]
 [ 98  11]]
```

```
y_pred_test_svc=svc.predict(x_test)
```

```
b2=accuracy_score(y_test,y_pred_test_svc)
b2
```

```
0.7586206896551724
```

```
print(classification_report(y_test,y_pred_test_svc))
```

```
              precision    recall  f1-score   support

           0       0.76      1.00      0.86        87
           1       1.00      0.03      0.07        29

    accuracy                           0.76       116
   macro avg       0.88      0.52      0.46       116
weighted avg       0.82      0.76      0.66       116
```

```
print(confusion_matrix(y_test,y_pred_test_svc))
```

```
[[87  0]
 [28  1]]
```

```
svc1=SVC()
svc1.fit(x_sm,y_sm)
```

```
SVC()

y_pred=svc1.predict(x_test)
c2=accuracy_score(y_test,y_pred)
c2

0.6724137931034483
```

## Hyperparameter Tunnung

```
params={'C':range(1,1000,10),'gamma': [1, 0.1, 0.01, 0.001, 0.0001],
'kernel': ['linear','poly','rbf','sigmoid']}
svc=RandomizedSearchCV(SVC(),param_distributions=params,verbose=2,rand
om_state=42)
svc.fit(x_train, y_train)

Fitting 5 folds for each of 10 candidates, totalling 50 fits
[CV] END .....................C=561, gamma=0.1, kernel=rbf; total
time=   0.0s
[CV] END .....................C=561, gamma=0.1, kernel=rbf; total
time=   0.0s
[CV] END .....................C=561, gamma=0.1, kernel=rbf; total
time=   0.0s
[CV] END .....................C=561, gamma=0.1, kernel=rbf; total
time=   0.0s
[CV] END .....................C=561, gamma=0.1, kernel=rbf; total
time=   0.0s
[CV] END ...............C=721, gamma=0.0001, kernel=sigmoid; total
time=   0.0s
[CV] END ...............C=721, gamma=0.0001, kernel=sigmoid; total
time=   0.0s
[CV] END ...............C=721, gamma=0.0001, kernel=sigmoid; total
time=   0.0s
[CV] END ...............C=721, gamma=0.0001, kernel=sigmoid; total
time=   0.0s
[CV] END ...............C=721, gamma=0.0001, kernel=sigmoid; total
time=   0.0s
[CV] END ....................C=431, gamma=1, kernel=linear; total
time=   2.3s
[CV] END ....................C=431, gamma=1, kernel=linear; total
time=   3.3s
[CV] END ....................C=431, gamma=1, kernel=linear; total
time=   1.2s
[CV] END ....................C=431, gamma=1, kernel=linear; total
time=   3.9s
[CV] END ....................C=431, gamma=1, kernel=linear; total
time=   3.9s
[CV] END ....................C=641, gamma=0.001, kernel=rbf; total
time=   0.0s
[CV] END ....................C=641, gamma=0.001, kernel=rbf; total
```

```
time=   0.0s
[CV] END .....................C=641, gamma=0.001, kernel=rbf; total
time=   0.0s
[CV] END .....................C=641, gamma=0.001, kernel=rbf; total
time=   0.0s
[CV] END .....................C=641, gamma=0.001, kernel=rbf; total
time=   0.0s
[CV] END ......................C=561, gamma=0.01, kernel=rbf; total
time=   0.0s
[CV] END ......................C=561, gamma=0.01, kernel=rbf; total
time=   0.0s
[CV] END ......................C=561, gamma=0.01, kernel=rbf; total
time=   0.0s
[CV] END ......................C=561, gamma=0.01, kernel=rbf; total
time=   0.0s
[CV] END ......................C=561, gamma=0.01, kernel=rbf; total
time=   0.0s
[CV] END .................C=541, gamma=0.001, kernel=sigmoid; total
time=   0.0s
[CV] END .................C=541, gamma=0.001, kernel=sigmoid; total
time=   0.0s
[CV] END .................C=541, gamma=0.001, kernel=sigmoid; total
time=   0.0s
[CV] END .................C=541, gamma=0.001, kernel=sigmoid; total
time=   0.0s
[CV] END .................C=541, gamma=0.001, kernel=sigmoid; total
time=   0.0s
[CV] END ....................C=861, gamma=0.1, kernel=linear; total
time=   5.2s
[CV] END ....................C=861, gamma=0.1, kernel=linear; total
time=   3.4s
[CV] END ....................C=861, gamma=0.1, kernel=linear; total
time=   5.0s
[CV] END ....................C=861, gamma=0.1, kernel=linear; total
time=   5.8s
[CV] END ....................C=861, gamma=0.1, kernel=linear; total
time=   9.5s
[CV] END ....................C=521, gamma=0.1, kernel=linear; total
time=   3.9s
[CV] END ....................C=521, gamma=0.1, kernel=linear; total
time=   3.7s
[CV] END ....................C=521, gamma=0.1, kernel=linear; total
time=   2.5s
[CV] END ....................C=521, gamma=0.1, kernel=linear; total
time=   3.6s
[CV] END ....................C=521, gamma=0.1, kernel=linear; total
time=   4.7s
[CV] END ....................C=811, gamma=0.0001, kernel=rbf; total
time=   0.0s
```

```
[CV] END ....................C=811, gamma=0.0001, kernel=rbf; total
time=   0.0s
[CV] END ....................C=811, gamma=0.0001, kernel=rbf; total
time=   0.0s
[CV] END ....................C=811, gamma=0.0001, kernel=rbf; total
time=   0.0s
[CV] END ....................C=811, gamma=0.0001, kernel=rbf; total
time=   0.0s
[CV] END .........................C=61, gamma=1, kernel=poly; total
time=   5.6s
[CV] END .........................C=61, gamma=1, kernel=poly; total
time=   9.5s
[CV] END .........................C=61, gamma=1, kernel=poly; total
time=   3.8s
[CV] END .........................C=61, gamma=1, kernel=poly; total
time=  14.2s
[CV] END .........................C=61, gamma=1, kernel=poly; total
time=  13.4s
```

```
RandomizedSearchCV(estimator=SVC(),
                   param_distributions={'C': range(1, 1000, 10),
                                        'gamma': [1, 0.1, 0.01, 0.001,
0.0001],
                                        'kernel': ['linear', 'poly',
'rbf',
                                                   'sigmoid']},
                   random_state=42, verbose=2)
```

```
d2=svc.best_params_
d2
```

```
{'kernel': 'rbf', 'gamma': 0.001, 'C': 641}
```

```
svc_classifier=SVC(kernel='rbf',gamma= 0.001,C=641)
svc_classifier.fit(x_train,y_train)
print(svc_classifier.score(x_test,y_test))
```

```
0.75
```

```
y_predict = svc.predict(x_test)
```

```
e2=accuracy_score(y_test,y_predict)
e2
```

```
0.75
```

```
print(classification_report(y_test, y_predict))
```

```
              precision    recall  f1-score   support

           0       0.75      1.00      0.86        87
           1       0.00      0.00      0.00        29
```

```
    accuracy                           0.75       116
   macro avg       0.38      0.50      0.43       116
weighted avg       0.56      0.75      0.64       116
```

## Decision Tree

```
from sklearn.tree import DecisionTreeClassifier

DT=DecisionTreeClassifier(criterion='gini',
    splitter='best',
    max_depth=None,
    min_samples_split=2,
    min_samples_leaf=1,
    min_weight_fraction_leaf=0.0,
    max_features=None,
    random_state=None,
    max_leaf_nodes=None,
    min_impurity_decrease=0.0,
    class_weight=None,
    ccp_alpha=0.0)

DT.fit(x_train,y_train)

DecisionTreeClassifier()

y_pred_train_dt=DT.predict(x_train)

a3=accuracy_score(y_train,y_pred_train_dt)
a3

0.9434782608695652

y_pred_test_dt=DT.predict(x_test)

b3=accuracy_score(y_test,y_pred_test_dt)
b3

0.6724137931034483

dt1=DecisionTreeClassifier()

dt1.fit(x_sm,y_sm)

DecisionTreeClassifier()

y_pred=dt1.predict(x_test)

c3=accuracy_score(y_test,y_pred)
c3
```

```
0.646551724137931
```

## Hyperparameter Tunnung

```
from sklearn.model_selection import GridSearchCV

#creating dictionary--> key value pair of hyperparameters having key
as parameter and values as its values
params = {
    "criterion":("gini", "entropy"), #quality of split
    "splitter":("best", "random"), # searches the features for a split
    "max_depth":(list(range(1, 20))), #depth of tree range from 1 to
19
    "min_samples_split":[2, 3, 4],    #the minimum number of samples
required to split internal node
    "min_samples_leaf":list(range(1, 20)),#minimum number of samples
required to be at a leaf node,we are passing list which is range from
1 to 19
}


tree_clf = DecisionTreeClassifier(random_state=3)#object creation for
decision tree with random state 3
tree_cv = GridSearchCV(tree_clf, params, scoring="f1", n_jobs=-1,
verbose=1, cv=3)
#passing model to gridsearchCV ,
#tree_clf-->model
#params---->hyperparametes(dictionary we created)
#scoring--->performance matrix to check performance
#n_jobs---->Number of jobs to run in parallel,-1 means using all
processors.
#verbose=Controls the verbosity: the higher, the more messages.
#>1 : the computation time for each fold and parameter candidate is
displayed;
#>2 : the score is also displayed;
#>3 : the fold and candidate parameter indexes are also displayed
together with the starting time of the computation.
#cv------> number of flods




tree_cv.fit(x_train,y_train)#training data on gridsearch cv
best_params = tree_cv.best_params_#it will give you best parameters
print(f"Best paramters: {best_params})")#printing  best parameters
```

```
Fitting 3 folds for each of 4332 candidates, totalling 12996 fits
Best paramters: {'criterion': 'entropy', 'max_depth': 7,
'min_samples_leaf': 6, 'min_samples_split': 2, 'splitter': 'best'})
```

```
Best_parameters:({'criterion': 'gini', 'max_depth': 5,
'min_samples_leaf': 7, 'min_samples_split': 2, 'splitter': 'best'})
```

```
d3=tree_cv.best_params_
d3
```

```
{'criterion': 'entropy',
 'max_depth': 7,
 'min_samples_leaf': 6,
 'min_samples_split': 2,
 'splitter': 'best'}
```

```
tree_cv.best_score_
```

```
0.4679738562091503
```

```
dt2=DecisionTreeClassifier(criterion='entropy', max_depth= 7,
min_samples_leaf= 6, min_samples_split= 2, splitter= 'best')
```

```
dt2.fit(x_train,y_train)
```

```
DecisionTreeClassifier(criterion='entropy', max_depth=7,
min_samples_leaf=6)
```

```
y_pred=dt2.predict(x_test)
```

```
e3=accuracy_score(y_test,y_pred)
e3
```

```
0.7758620689655172
```

```
print(classification_report(y_test,y_pred))
```

```
              precision    recall  f1-score   support

           0       0.88      0.82      0.85        87
           1       0.54      0.66      0.59        29

    accuracy                           0.78       116
   macro avg       0.71      0.74      0.72       116
weighted avg       0.79      0.78      0.78       116
```

# KNN

```
from sklearn.neighbors import KNeighborsClassifier
knn=KNeighborsClassifier()
```

```
ERROR_RATE = []
```

```
for i in range(1,13):
    knn=KNeighborsClassifier(n_neighbors=i)
```

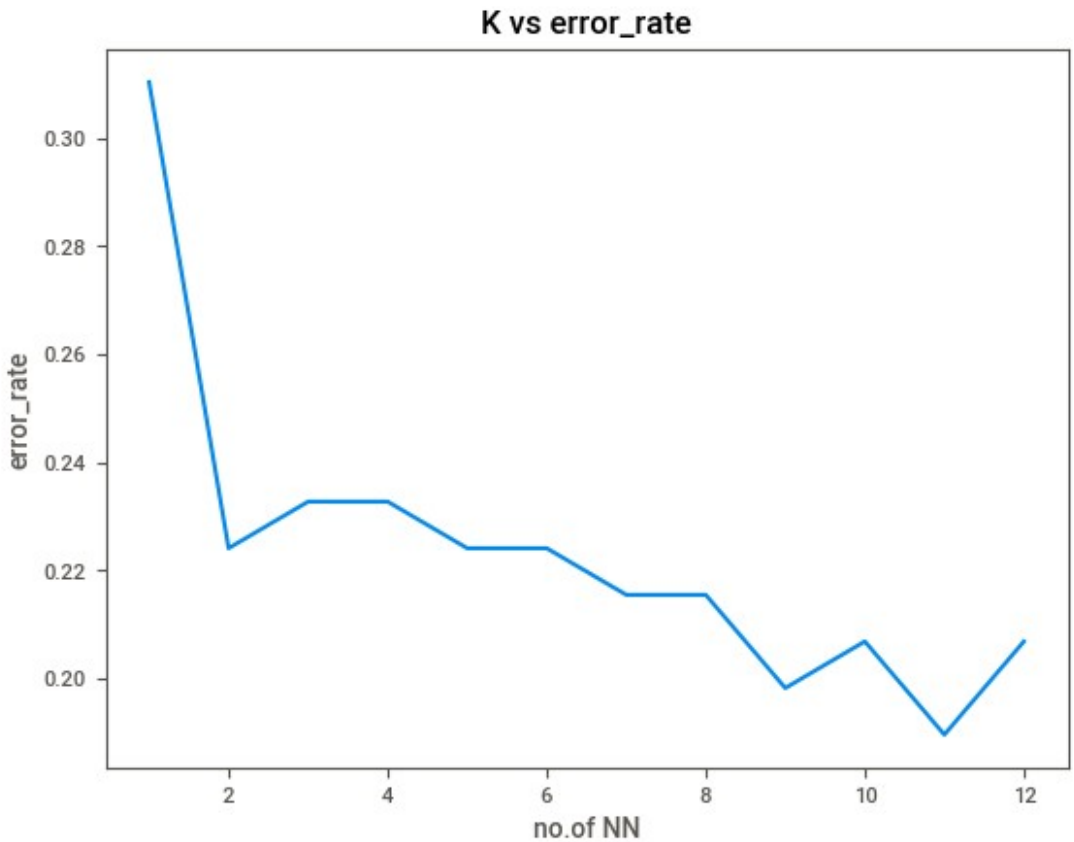
```
    knn.fit(x_train,y_train)
    y_pred=knn.predict(x_test)
    error_rate=(y_test != y_pred).sum()/len(y_test)
    ERROR_RATE.append(error_rate)
```

```
ERROR_RATE
```

```
[0.3103448275862069,
 0.22413793103448276,
 0.23275862068965517,
 0.23275862068965517,
 0.22413793103448276,
 0.22413793103448276,
 0.21551724137931033,
 0.21551724137931033,
 0.19827586206896552,
 0.20689655172413793,
 0.1896551724137931,
 0.20689655172413793]
```

```
plt.plot(range(1,13),ERROR_RATE)
plt.xlabel('no.of NN')
plt.ylabel('error_rate')
plt.title('K vs error_rate')
```

```
Text(0.5, 1.0, 'K vs error_rate')
```




```
knn1=KNeighborsClassifier(n_neighbors=7)

knn1.fit(x_train,y_train)

KNeighborsClassifier(n_neighbors=7)

y_pred_train_knn=knn1.predict(x_train)

a4=accuracy_score(y_train,y_pred_train_knn)
a4

0.8260869565217391

y_pred_test_knn=knn1.predict(x_test)

b4=accuracy_score(y_test,y_pred_test_knn)
b4

0.7844827586206896

knn2=KNeighborsClassifier()
knn2.fit(x_sm,y_sm)

KNeighborsClassifier()
```

```
y_pred=knn2.predict(x_test)

c4=accuracy_score(y_test,y_pred)
c4
```

```
0.7672413793103449
```

## Hyperparameter Tunnung

```
params={'n_neighbors' : range(1,40),'weights' :
['uniform','distance'],'metric' :
['minkowski','euclidean','manhattan']}
knn=RandomizedSearchCV(KNeighborsClassifier(),param_distributions=para
ms, verbose=2, cv=3, n_jobs = -1, random_state=42)
knn.fit(x_train, y_train)
```

```
Fitting 3 folds for each of 10 candidates, totalling 30 fits

RandomizedSearchCV(cv=3, estimator=KNeighborsClassifier(), n_jobs=-1,
                   param_distributions={'metric': ['minkowski',
'euclidean',
                                                   'manhattan'],
                                        'n_neighbors': range(1, 40),
                                        'weights': ['uniform',
'distance']},
                   random_state=42, verbose=2)
```

```
d4=knn.best_params_
d4
```

```
{'weights': 'distance', 'n_neighbors': 25, 'metric': 'euclidean'}
```

```
print(knn.best_estimator_)
```

```
KNeighborsClassifier(metric='euclidean', n_neighbors=25,
weights='distance')
```

```
knn_classifier=KNeighborsClassifier(weights='distance',n_neighbors= 25
,metric='euclidean')
knn_classifier.fit(x_train,y_train)
print(knn_classifier.score(x_test,y_test))
```

```
0.7758620689655172
```

```
knn_y_pred = knn.predict(x_test)
e4=accuracy_score(y_test,knn_y_pred)
e4
```

```
0.7758620689655172
```

```
# print classification report
print(classification_report(y_test, knn_y_pred))
```

```
              precision    recall  f1-score   support

           0       0.83      0.89      0.86        87
           1       0.57      0.45      0.50        29

    accuracy                           0.78       116
   macro avg       0.70      0.67      0.68       116
weighted avg       0.76      0.78      0.77       116
```

# XGB

```
from xgboost import XGBClassifier

xgb=XGBClassifier()
xgb.fit(x_train,y_train)
```

```
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None,
early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None,
feature_types=None,
              gamma=None, grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=None,
max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None, max_leaves=None,
              min_child_weight=None, missing=nan,
monotone_constraints=None,
              multi_strategy=None, n_estimators=None, n_jobs=None,
              num_parallel_tree=None, random_state=None, ...)
```

```
y_pred_train_xgb=xgb.predict(x_train)

a5=accuracy_score(y_train,y_pred_train_xgb)
a5
```

```
0.9326086956521739
```

```
y_pred_test_xgb=xgb.predict(x_test)

b5=accuracy_score(y_test,y_pred_test_xgb)
b5
```

```
0.7413793103448276
```

```
xgb1=XGBClassifier()
xgb1.fit(x_sm,y_sm)
```

```
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None,
early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None,
feature_types=None,
              gamma=None, grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=None,
max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=None, max_leaves=None,
              min_child_weight=None, missing=nan,
monotone_constraints=None,
              multi_strategy=None, n_estimators=None, n_jobs=None,
              num_parallel_tree=None, random_state=None, ...)
```

```
y_pred=xgb1.predict(x_test)
```

```
c5=accuracy_score(y_test,y_pred)
c5
```

```
0.75
```

## Hyperparameter Tunnung

```
param_grid={
    'max_depth':[3,5,7],
    'learning_rate':[0.01,0.1,0.3],
    'n_estimators':[100,200,300],
    'subsample':[0.8,0.9,1.0],
    'colsample_bytree':[0.8,0.9,1.0],
    'gamma':[0.1,0.2,0.3],
    'reg_alpha':[0.1,0.2,0.3],
    'reg_lambda':[0.1,0.2,0.3]
}
```

```
grid_search=GridSearchCV(estimator=xgb,param_grid=param_grid,cv=5,scor
ing='accuracy',n_jobs=-1)
```

```
grid_search.fit(x_train,y_train)
```

```
GridSearchCV(cv=5,
             estimator=XGBClassifier(base_score=None, booster=None,
                                     callbacks=None,
colsample_bylevel=None,
                                     colsample_bynode=None,
                                     colsample_bytree=None,
device=None,
                                     early_stopping_rounds=None,
                                     enable_categorical=False,
eval_metric=None,
```

```
                                        feature_types=None, gamma=None,
                                        grow_policy=None,
importance_type=None,
                                        interaction_constraints=None,
                                        learning_rate=None,...
                                        multi_strategy=None,
n_estimators=None,
                                        n_jobs=None,
num_parallel_tree=None,
                                        random_state=None, ...),
             n_jobs=-1,
             param_grid={'colsample_bytree': [0.8, 0.9, 1.0],
                         'gamma': [0.1, 0.2, 0.3],
                         'learning_rate': [0.01, 0.1, 0.3],
                         'max_depth': [3, 5, 7],
                         'n_estimators': [100, 200, 300],
                         'reg_alpha': [0.1, 0.2, 0.3],
                         'reg_lambda': [0.1, 0.2, 0.3],
                         'subsample': [0.8, 0.9, 1.0]},
             scoring='accuracy')
```

```
d5=grid_search.best_params_
d5
```

```
{'colsample_bytree': 1.0,
 'gamma': 0.2,
 'learning_rate': 0.01,
 'max_depth': 7,
 'n_estimators': 100,
 'reg_alpha': 0.3,
 'reg_lambda': 0.3,
 'subsample': 0.8}
```

```
grid_search.best_score_
```

```
0.7804347826086956
```

```
grid_search1=XGBClassifier(colsample_bytree = 1.0,gamma=
0.2,learning_rate= 0.01,max_depth= 7,n_estimators=100,reg_alpha=
0.3,reg_lambda= 0.3,subsample=0.8)
```

```
grid_search1.fit(x_train,y_train)
```

```
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=1.0, device=None,
early_stopping_rounds=None,
              enable_categorical=False, eval_metric=None,
feature_types=None,
              gamma=0.2, grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=0.01,
```

```
max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=7, max_leaves=None,
              min_child_weight=None, missing=nan,
monotone_constraints=None,
              multi_strategy=None, n_estimators=100, n_jobs=None,
              num_parallel_tree=None, random_state=None, ...)
```

```
y_pred=grid_search1.predict(x_test)
```

```
e5=accuracy_score(y_test,y_pred)
e5
```

```
0.7758620689655172
```

```
predictions=pd.DataFrame({'model':['Random Forest','SVM','Decision
Tree','KNN','XGB'] ,
                        'accuracy_score_train':[a1,a2,a3,a4,a5],
                      'accuracy_score_test':[b1,b2,b3,b4,b5],
                         'accuracy_score_smote_test':
[c1,c2,c3,c4,c5]})
```

```
predictions
```

```
           model  accuracy_score_train  accuracy_score_test  \
0  Random Forest              0.943478             0.741379
1            SVM              0.780435             0.758621
2  Decision Tree              0.943478             0.672414
3            KNN              0.826087             0.784483
4            XGB              0.932609             0.741379

   accuracy_score_smote_test
0                   0.724138
1                   0.672414
2                   0.646552
3                   0.767241
4                   0.750000
```

```
hyperparameter_predictions=pd.DataFrame({'model':['Random
Forest','SVM','Decision Tree','KNN','XGB'] ,
                        'best params':[d1,d2,d3,d4,d5],
                      'accuracy_score_test':[e1,e2,e3,e4,e5],
                         })
```

```
hyperparameter_predictions
```

```
           model                                          best params  \
0  Random Forest  {'n_estimators': 1000, 'min_samples_split': 10...
1            SVM        {'kernel': 'rbf', 'gamma': 0.001, 'C': 641}
2  Decision Tree  {'criterion': 'entropy', 'max_depth': 7, 'min_...
3            KNN  {'weights': 'distance', 'n_neighbors': 25, 'me...
```

```
4            XGB  {'colsample_bytree': 1.0, 'gamma': 0.2, 'learn...

   accuracy_score_test
0             0.750000
1             0.750000
2             0.775862
3             0.775862
4             0.775862

[CV] END bootstrap=True, max_depth=30, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=400; total time=
0.3s
[CV] END bootstrap=False, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1200; total
time=   0.7s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=800; total time=
0.4s
[CV] END bootstrap=False, max_depth=100, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1000; total
time=   0.6s
[CV] END bootstrap=False, max_depth=90, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=800; total time=
0.5s
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min_samples_leaf=2, min_samples_split=10, n_estimators=800; total
time=   0.5s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=800; total
time=   0.5s
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min_samples_leaf=1, min_samples_split=5, n_estimators=1600; total
time=   1.0s
[CV] END bootstrap=False, max_depth=None, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=400; total time=
0.3s
[CV] END bootstrap=False, max_depth=None, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=400; total time=
0.3s
[CV] END bootstrap=False, max_depth=40, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1400; total
time=   0.0s
[CV] END bootstrap=False, max_depth=40, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1400; total
time=   0.0s
[CV] END bootstrap=False, max_depth=40, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1400; total
time=   0.0s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1000; total
```

```
time=   0.9s
[CV] END bootstrap=False, max_depth=50, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=800; total time=
0.5s
[CV] END bootstrap=True, max_depth=50, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=800; total
time=   0.7s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1200; total
time=   1.0s
[CV] END bootstrap=True, max_depth=60, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=600; total
time=   0.6s
[CV] END bootstrap=False, max_depth=90, max_features=auto,
min_samples_leaf=2, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=False, max_depth=90, max_features=auto,
min_samples_leaf=2, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=False, max_depth=90, max_features=auto,
min_samples_leaf=2, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=True, max_depth=110, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
time=   0.9s
[CV] END bootstrap=False, max_depth=110, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=600; total time=
0.4s
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min_samples_leaf=2, min_samples_split=10, n_estimators=1400; total
time=   1.1s
[CV] END bootstrap=True, max_depth=80, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=1400; total
time=   1.0s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1000; total
time=   0.8s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1000; total
time=   0.8s
[CV] END bootstrap=True, max_depth=100, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=400; total time=
0.3s
[CV] END bootstrap=True, max_depth=100, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=400; total time=
0.4s
[CV] END bootstrap=True, max_depth=110, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=600; total
time=   0.5s
```

```
[CV] END bootstrap=True, max_depth=110, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=600; total
time=   0.5s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=600; total time=
0.3s
[CV] END bootstrap=False, max_depth=40, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=400; total
time=   0.0s
[CV] END bootstrap=False, max_depth=40, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=400; total
time=   0.0s
[CV] END bootstrap=False, max_depth=40, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=400; total
time=   0.0s
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time=   0.0s
[CV] END bootstrap=False, max_depth=10, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=10, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=100, max_features=auto,
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0.0s
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min_samples_leaf=4, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=False, max_depth=100, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=2000; total
time=   1.2s
[CV] END .metric=minkowski, n_neighbors=35, weights=distance; total
time=   0.0s
[CV] END ..metric=manhattan, n_neighbors=13, weights=uniform; total
time=   0.0s
[CV] END ..metric=minkowski, n_neighbors=5, weights=distance; total
time=   0.0s
[CV] END .metric=euclidean, n_neighbors=25, weights=distance; total
time=   0.0s
[CV] END ..metric=manhattan, n_neighbors=21, weights=uniform; total
time=   0.0s
[CV] END ..metric=euclidean, n_neighbors=8, weights=distance; total
time=   0.0s
[CV] END ..metric=minkowski, n_neighbors=8, weights=distance; total
```

```
time=   0.0s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=2000; total
time=   1.5s
[CV] END bootstrap=False, max_depth=60, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=600; total time=
0.3s
[CV] END bootstrap=False, max_depth=10, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=2000; total
time=   1.2s
[CV] END bootstrap=False, max_depth=110, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=110, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=1800; total
time=   0.0s
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min_samples_leaf=2, min_samples_split=10, n_estimators=1800; total
time=   0.0s
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min_samples_leaf=1, min_samples_split=5, n_estimators=600; total time=
0.0s
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0.0s
[CV] END bootstrap=True, max_depth=80, max_features=auto,
min_samples_leaf=1, min_samples_split=5, n_estimators=600; total time=
0.0s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1800; total
time=   1.0s
[CV] END bootstrap=True, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=1400; total
time=   1.2s
[CV] END bootstrap=False, max_depth=20, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=1200; total
time=   0.7s
[CV] END bootstrap=False, max_depth=50, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=800; total time=
0.6s
[CV] END bootstrap=True, max_depth=90, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1800; total
time=   1.6s
[CV] END bootstrap=True, max_depth=90, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=800; total time=
0.7s
[CV] END bootstrap=True, max_depth=110, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
time=   0.8s
```

```
[CV] END bootstrap=False, max_depth=None, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1400; total
time=   0.9s
[CV] END bootstrap=False, max_depth=60, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=60, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1400; total
time=   0.8s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1400; total
time=   0.7s
[CV] END bootstrap=True, max_depth=100, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=2000; total
time=   1.4s
[CV] END bootstrap=True, max_depth=100, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=2000; total
time=   1.5s
[CV] END bootstrap=False, max_depth=None, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=2000; total
time=   1.1s
[CV] END bootstrap=False, max_depth=None, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=2000; total
time=   0.8s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
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time=   1.5s
[CV] END bootstrap=False, max_depth=60, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=600; total time=
0.3s
[CV] END bootstrap=False, max_depth=10, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=2000; total
time=   1.3s
[CV] END bootstrap=False, max_depth=70, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=1600; total
time=   0.9s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=1400; total
time=   0.9s
[CV] END bootstrap=False, max_depth=100, max_features=auto,
min_samples_leaf=4, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=100, max_features=auto,
min_samples_leaf=4, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=100, max_features=auto,
```

```
min_samples_leaf=4, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=1200; total
time=   0.9s
[CV] END bootstrap=False, max_depth=100, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=800; total time=
0.5s
[CV] END bootstrap=True, max_depth=90, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1800; total
time=   1.5s
[CV] END bootstrap=True, max_depth=10, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=True, max_depth=10, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=True, max_depth=10, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=True, max_depth=60, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=600; total
time=   0.5s
[CV] END bootstrap=True, max_depth=90, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=400; total
time=   0.3s
[CV] END bootstrap=True, max_depth=90, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=True, max_depth=90, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=True, max_depth=90, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=400; total
time=   0.3s
[CV] END bootstrap=False, max_depth=110, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=600; total time=
0.4s
[CV] END bootstrap=False, max_depth=20, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=20, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=20, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1800; total
```

```
time=   0.0s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=4, min_samples_split=10, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=4, min_samples_split=10, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=4, min_samples_split=10, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=30, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=30, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=30, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=None, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1400; total
time=   0.9s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1600; total
time=   1.2s
[CV] END bootstrap=True, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=1600; total
time=   0.0s
[CV] END bootstrap=True, max_depth=60, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1800; total
time=   1.2s
[CV] END bootstrap=False, max_depth=90, max_features=auto,
min_samples_leaf=1, min_samples_split=5, n_estimators=400; total time=
0.0s
[CV] END bootstrap=False, max_depth=70, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=800; total time=
0.5s
[CV] END bootstrap=False, max_depth=40, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=600; total
time=   0.3s
```

```
[CV] END bootstrap=True, max_depth=100, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=800; total
time=   0.7s
[CV] END bootstrap=True, max_depth=100, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=800; total
time=   0.6s
[CV] END bootstrap=True, max_depth=40, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=800; total time=
0.6s
[CV] END bootstrap=True, max_depth=40, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=800; total time=
0.6s
[CV] END .metric=minkowski, n_neighbors=35, weights=distance; total
time=   0.0s
[CV] END ..metric=manhattan, n_neighbors=26, weights=uniform; total
time=   0.0s
[CV] END ..metric=minkowski, n_neighbors=5, weights=distance; total
time=   0.0s
[CV] END .metric=euclidean, n_neighbors=25, weights=distance; total
time=   0.0s
[CV] END .metric=euclidean, n_neighbors=16, weights=distance; total
time=   0.0s
[CV] END ..metric=euclidean, n_neighbors=8, weights=distance; total
time=   0.0s
[CV] END ..metric=euclidean, n_neighbors=8, weights=distance; total
time=   0.0s
[CV] END ..metric=minkowski, n_neighbors=8, weights=distance; total
time=   0.0s
[CV] END bootstrap=True, max_depth=30, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=400; total time=
0.3s
[CV] END bootstrap=False, max_depth=30, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=30, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=30, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1600; total
time=   1.1s
[CV] END bootstrap=False, max_depth=100, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1000; total
time=   0.6s
[CV] END bootstrap=False, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1600; total
time=   1.0s
```

```
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
time=   0.8s
[CV] END bootstrap=True, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=1400; total
time=   1.1s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1000; total
time=   0.9s
[CV] END bootstrap=False, max_depth=50, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=800; total time=
0.5s
[CV] END bootstrap=True, max_depth=50, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=800; total
time=   0.7s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1200; total
time=   1.3s
[CV] END bootstrap=True, max_depth=90, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=800; total time=
0.7s
[CV] END bootstrap=False, max_depth=70, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1200; total
time=   0.8s
[CV] END bootstrap=True, max_depth=10, max_features=auto,
min_samples_leaf=1, min_samples_split=5, n_estimators=1600; total
time=   0.0s
[CV] END bootstrap=True, max_depth=10, max_features=auto,
min_samples_leaf=1, min_samples_split=5, n_estimators=1600; total
time=   0.0s
[CV] END bootstrap=True, max_depth=10, max_features=auto,
min_samples_leaf=1, min_samples_split=5, n_estimators=1600; total
time=   0.0s
[CV] END bootstrap=True, max_depth=80, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=True, max_depth=80, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=True, max_depth=80, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1400; total
time=   0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1400; total
time=   0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
```

```
min_samples_leaf=1, min_samples_split=2, n_estimators=1400; total
time=   0.0s
[CV] END bootstrap=True, max_depth=80, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=1400; total
time=   1.1s
[CV] END bootstrap=False, max_depth=20, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=800; total time=
0.4s
[CV] END bootstrap=False, max_depth=20, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=800; total time=
0.5s
[CV] END bootstrap=False, max_depth=90, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=400; total
time=   0.3s
[CV] END bootstrap=False, max_depth=90, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=400; total
time=   0.2s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1000; total
time=   0.7s
[CV] END bootstrap=True, max_depth=80, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=False, max_depth=40, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=600; total
time=   0.3s
[CV] END bootstrap=False, max_depth=40, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=600; total
time=   0.3s
[CV] END bootstrap=True, max_depth=100, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=800; total
time=   0.6s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=1000; total
time=   0.8s
[CV] END bootstrap=True, max_depth=40, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=800; total time=
0.6s
[CV] END bootstrap=True, max_depth=40, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=600; total time=
0.0s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=2000; total
time=   1.2s
[CV] END .metric=minkowski, n_neighbors=35, weights=distance; total
time=   0.0s
[CV] END ..metric=manhattan, n_neighbors=26, weights=uniform; total
time=   0.0s
[CV] END ..metric=manhattan, n_neighbors=13, weights=uniform; total
```

```
time=   0.0s
[CV] END .metric=euclidean, n_neighbors=16, weights=distance; total
time=   0.0s
[CV] END .metric=manhattan, n_neighbors=29, weights=distance; total
time=   0.0s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=2000; total
time=   1.5s
[CV] END bootstrap=False, max_depth=60, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=600; total time=
0.3s
[CV] END bootstrap=False, max_depth=90, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=800; total time=
0.5s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=800; total
time=   0.6s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=70, max_features=auto,
min_samples_leaf=2, min_samples_split=5, n_estimators=600; total time=
0.0s
[CV] END bootstrap=False, max_depth=70, max_features=auto,
min_samples_leaf=2, min_samples_split=5, n_estimators=600; total time=
0.0s
[CV] END bootstrap=False, max_depth=70, max_features=auto,
min_samples_leaf=2, min_samples_split=5, n_estimators=600; total time=
0.0s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
time=   0.8s
[CV] END bootstrap=False, max_depth=70, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=1600; total
time=   1.1s
[CV] END bootstrap=False, max_depth=None, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=400; total time=
0.3s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=1200; total
time=   1.1s
[CV] END bootstrap=False, max_depth=100, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=800; total time=
0.6s
```

```
[CV] END bootstrap=False, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=800; total
time=   0.5s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=800; total time=
0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=800; total time=
0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=800; total time=
0.0s
[CV] END bootstrap=True, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=5, n_estimators=800; total time=
0.0s
[CV] END bootstrap=True, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=5, n_estimators=800; total time=
0.0s
[CV] END bootstrap=True, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=5, n_estimators=800; total time=
0.0s
[CV] END bootstrap=True, max_depth=60, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=1000; total
time=   0.8s
[CV] END bootstrap=True, max_depth=60, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=600; total
time=   0.5s
[CV] END bootstrap=True, max_depth=90, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=400; total
time=   0.3s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=400; total
time=   0.3s
[CV] END bootstrap=False, max_depth=110, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=600; total time=
0.4s
[CV] END bootstrap=False, max_depth=60, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=600; total time=
0.4s
[CV] END bootstrap=True, max_depth=50, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=200; total time=
0.2s
[CV] END bootstrap=True, max_depth=50, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=200; total time=
0.2s
[CV] END bootstrap=False, max_depth=80, max_features=auto,
min_samples_leaf=4, min_samples_split=10, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=80, max_features=auto,
```

```
min_samples_leaf=4, min_samples_split=10, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=80, max_features=auto,
min_samples_leaf=4, min_samples_split=10, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=1200; total
time=   0.0s
[CV] END bootstrap=False, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=1200; total
time=   0.0s
[CV] END bootstrap=False, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=1200; total
time=   0.0s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1600; total
time=   1.2s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1600; total
time=   1.3s
[CV] END bootstrap=True, max_depth=80, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=True, max_depth=80, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=200; total time=
0.0s
[CV] END bootstrap=False, max_depth=60, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=60, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=60, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=600; total
time=   0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=600; total
time=   0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=600; total
time=   0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total
```

```
time=   0.0s
[CV] END bootstrap=True, max_depth=100, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=70, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=800; total time=
0.4s
[CV] END bootstrap=False, max_depth=70, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=800; total time=
0.4s
[CV] END bootstrap=True, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1600; total
time=   1.2s
[CV] END bootstrap=True, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1600; total
time=   1.2s
[CV] END ..metric=manhattan, n_neighbors=26, weights=uniform; total
time=   0.0s
[CV] END ..metric=manhattan, n_neighbors=13, weights=uniform; total
time=   0.0s
[CV] END .metric=euclidean, n_neighbors=16, weights=distance; total
time=   0.0s
[CV] END ..metric=manhattan, n_neighbors=21, weights=uniform; total
time=   0.0s
[CV] END .metric=manhattan, n_neighbors=29, weights=distance; total
time=   0.0s
[CV] END bootstrap=False, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1200; total
time=   0.7s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=800; total time=
0.4s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=800; total time=
0.5s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
```

```
time=   0.0s
[CV] END bootstrap=False, max_depth=50, max_features=auto,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
time=   0.0s
[CV] END bootstrap=False, max_depth=10, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=10, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=10, max_features=auto,
min_samples_leaf=4, min_samples_split=5, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=True, max_depth=70, max_features=auto,
min_samples_leaf=4, min_samples_split=10, n_estimators=400; total
time=   0.0s
[CV] END bootstrap=True, max_depth=70, max_features=auto,
min_samples_leaf=4, min_samples_split=10, n_estimators=400; total
time=   0.0s
[CV] END bootstrap=True, max_depth=70, max_features=auto,
min_samples_leaf=4, min_samples_split=10, n_estimators=400; total
time=   0.0s
[CV] END bootstrap=False, max_depth=90, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=800; total time=
0.4s
[CV] END bootstrap=False, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1600; total
time=   1.0s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1800; total
time=   1.0s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=1400; total
time=   1.1s
[CV] END bootstrap=False, max_depth=20, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=1200; total
time=   0.7s
[CV] END bootstrap=False, max_depth=100, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=800; total time=
0.5s
[CV] END bootstrap=True, max_depth=90, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1800; total
time=   1.5s
[CV] END bootstrap=True, max_depth=60, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=1000; total
time=   0.9s
[CV] END bootstrap=False, max_depth=70, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1200; total
time=   0.7s
```

```
[CV] END bootstrap=False, max_depth=60, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=600; total time=
0.4s
[CV] END bootstrap=True, max_depth=50, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=200; total time=
0.2s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=400; total time=
0.3s
[CV] END bootstrap=False, max_depth=10, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=True, max_depth=70, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=1400; total
time=   0.0s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1000; total
time=   0.6s
[CV] END bootstrap=False, max_depth=60, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=True, max_depth=60, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1800; total
time=   1.3s
[CV] END bootstrap=True, max_depth=60, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1800; total
time=   1.3s
[CV] END bootstrap=True, max_depth=100, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=400; total time=
0.3s
[CV] END bootstrap=True, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1600; total
time=   1.2s
[CV] END bootstrap=True, max_depth=40, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=600; total time=
0.0s
[CV] END bootstrap=True, max_depth=40, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=600; total time=
0.0s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=600; total time=
0.3s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=600; total time=
0.3s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=2000; total
time=   1.1s
[CV] END ..metric=minkowski, n_neighbors=8, weights=distance; total
```

```
time=   0.0s
[CV] END bootstrap=True, max_depth=30, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=400; total time=
0.3s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1600; total
time=   1.1s
[CV] END bootstrap=False, max_depth=100, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1000; total
time=   0.6s
[CV] END bootstrap=False, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1600; total
time=   1.0s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
time=   0.8s
[CV] END bootstrap=True, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=1400; total
time=   1.2s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1000; total
time=   0.8s
[CV] END bootstrap=False, max_depth=20, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=1200; total
time=   0.8s
[CV] END bootstrap=False, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=800; total
time=   0.5s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1200; total
time=   1.0s
[CV] END bootstrap=True, max_depth=90, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=800; total time=
0.7s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=400; total
time=   0.2s
[CV] END bootstrap=False, max_depth=70, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1200; total
time=   0.7s
[CV] END bootstrap=False, max_depth=60, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=600; total time=
0.4s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=400; total time=
0.3s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=400; total time=
0.3s
```

```
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1400; total
time=   0.8s
[CV] END bootstrap=False, max_depth=20, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=800; total time=
0.4s
[CV] END bootstrap=True, max_depth=None, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1600; total
time=   1.1s
[CV] END bootstrap=True, max_depth=100, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=2000; total
time=   1.5s
[CV] END bootstrap=True, max_depth=110, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=600; total
time=   0.5s
[CV] END bootstrap=False, max_depth=None, max_features=sqrt,
min_samples_leaf=4, min_samples_split=5, n_estimators=2000; total
time=   1.1s
[CV] END ..metric=minkowski, n_neighbors=5, weights=distance; total
time=   0.0s
[CV] END .metric=euclidean, n_neighbors=25, weights=distance; total
time=   0.0s
[CV] END ..metric=manhattan, n_neighbors=21, weights=uniform; total
time=   0.0s
[CV] END .metric=manhattan, n_neighbors=29, weights=distance; total
time=   0.0s
[CV] END bootstrap=False, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1200; total
time=   0.7s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=4, min_samples_split=2, n_estimators=1600; total
time=   1.2s
[CV] END bootstrap=False, max_depth=10, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=2000; total
time=   1.2s
[CV] END bootstrap=False, max_depth=30, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1800; total
time=   1.0s
[CV] END bootstrap=True, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=True, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=True, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=1, min_samples_split=5, n_estimators=1400; total
```

```
time=   1.0s
[CV] END bootstrap=True, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=1200; total
time=   1.1s
[CV] END bootstrap=True, max_depth=50, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=800; total
time=   0.6s
[CV] END bootstrap=False, max_depth=20, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=800; total
time=   0.6s
[CV] END bootstrap=True, max_depth=60, max_features=sqrt,
min_samples_leaf=2, min_samples_split=2, n_estimators=1000; total
time=   0.9s
[CV] END bootstrap=True, max_depth=90, max_features=sqrt,
min_samples_leaf=4, min_samples_split=10, n_estimators=400; total
time=   0.3s
[CV] END bootstrap=True, max_depth=110, max_features=sqrt,
min_samples_leaf=1, min_samples_split=2, n_estimators=1000; total
time=   0.9s
[CV] END bootstrap=False, max_depth=None, max_features=sqrt,
min_samples_leaf=2, min_samples_split=5, n_estimators=1400; total
time=   1.0s
[CV] END bootstrap=True, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=1600; total
time=   0.0s
[CV] END bootstrap=True, max_depth=None, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=1600; total
time=   0.0s
[CV] END bootstrap=False, max_depth=10, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=False, max_depth=10, max_features=auto,
min_samples_leaf=4, min_samples_split=2, n_estimators=1800; total
time=   0.0s
[CV] END bootstrap=True, max_depth=70, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=1400; total
time=   0.0s
[CV] END bootstrap=True, max_depth=70, max_features=auto,
min_samples_leaf=2, min_samples_split=2, n_estimators=1400; total
time=   0.0s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1000; total
time=   0.6s
[CV] END bootstrap=False, max_depth=80, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1000; total
time=   0.6s
[CV] END bootstrap=False, max_depth=90, max_features=auto,
min_samples_leaf=1, min_samples_split=5, n_estimators=400; total time=
0.0s
```

```
[CV] END bootstrap=False, max_depth=90, max_features=auto,
min_samples_leaf=1, min_samples_split=5, n_estimators=400; total time=
0.0s
[CV] END bootstrap=True, max_depth=90, max_features=auto,
min_samples_leaf=1, min_samples_split=5, n_estimators=1600; total
time=   0.0s
[CV] END bootstrap=True, max_depth=90, max_features=auto,
min_samples_leaf=1, min_samples_split=5, n_estimators=1600; total
time=   0.0s
[CV] END bootstrap=True, max_depth=90, max_features=auto,
min_samples_leaf=1, min_samples_split=5, n_estimators=1600; total
time=   0.0s
[CV] END bootstrap=False, max_depth=90, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=400; total
time=   0.2s
[CV] END bootstrap=True, max_depth=None, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1600; total
time=   1.1s
[CV] END bootstrap=True, max_depth=None, max_features=sqrt,
min_samples_leaf=1, min_samples_split=10, n_estimators=1600; total
time=   1.2s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=1000; total
time=   0.9s
[CV] END bootstrap=True, max_depth=10, max_features=sqrt,
min_samples_leaf=2, min_samples_split=10, n_estimators=1000; total
time=   0.8s
[CV] END bootstrap=True, max_depth=40, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=True, max_depth=40, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=2000; total
time=   0.0s
[CV] END bootstrap=True, max_depth=40, max_features=auto,
min_samples_leaf=2, min_samples_split=10, n_estimators=2000; total
time=   0.0s
```

From the above observation we can say that the performance of XGB model is good as compared to other models followed by Random Forest,SVM,Decision Tree, KNN. So we will save XGB model for further testing of the data using pickle library.

```
import pickle

file = open('blood_donation.pkl','wb')
pickle.dump(xgb,file)
```